

Technology Stack Specification

Date	06 July 2026	Project Name	Smart Lender
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Technology Stack Details			
S.No	Architecture Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Vanilla JS	Sleek light-mode dashboard matching bank styles, responsive tabs, async Fetch uploads without framework overhead.
2	Backend / Server-Side	Python, Flask framework	Fast, modular routing, lightweight runtime, perfect integration for running Python ML models.
3	Machine Learning Stack	scikit-learn, XGBoost, imbalanced-learn	High-performance classification, robust preprocessing, SMOTE class balancing, and StandardScaler scaling.
4	Data Storage / Data	CSV file formats, Pickle	In-memory spreadsheet parsing, local export files generation, and serialized pipeline parameters storage.
5	Version Control & CI/CD	Git, GitHub	Collaborative development, code quality tracking, branch deployments.
6	Cloud / Hosting	Render, Ngrok	Exposing local server publicly using secure tunnel for evaluator testing.